

# Sampling, de Branges, and the windowed Weil form

Where positivity lives; what the contrapositive does and does not say

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## Abstract

Windowed Weil positivity is a sampling inequality for the zeros of  $\zeta$  in the Paley–Wiener space  $PW_{L/2}$ . Under RH this is a theorem (*sampling-floor*, Theorem 1: Beurling through the explicit formula). Unconditionally the same quadratic form is Weil’s pairing, not a Gram of real samples: the contrapositive of *global* positivity is RH, but positivity on a single window  $W_L$  does not force every zero onto the line. An off-line zero produces a negative direction once the window is large enough to see it; below that scale the form can stay positive. The route does not prove RH. It says, in the language of Beurling density and de Branges spaces, where the positivity of  $Q_L$  lives, and which tools of that theory can enter.

## 1 The sampling reading, already a theorem

Fix  $L > 0$ ,  $\tau = L/2$ . Even  $f \in L^2([-L/2, L/2])$  have Fourier transforms  $F \in PW_\tau$ . Under RH the explicit formula on this window is

$$Q_L(f) = \sum_{\gamma} F(\gamma)^2,$$

so

$$c_L = \inf_{f \neq 0} \frac{Q_L(f)}{\|f\|^2}$$

is  $2\pi$  times the lower sampling constant of  $\Gamma = \{\pm\gamma\}$  for  $PW_\tau$ . Beurling–Landau: a separated subset  $\Lambda \subset \Gamma$  of lower uniform density  $D^-(\Lambda) > \tau/\pi$  is a sampling set. The zeros have  $N(T+1) - N(T) \rightarrow \infty$ , so such a  $\Lambda$  exists for every  $\tau$ . Hence  $c_L > 0$  and  $\lambda_{\min}(V_N) \downarrow c_L$  (*sampling-floor*, Theorem 1). Nothing further is claimed here.

The size of  $c_L$  is not a density statement. Density only gives  $c_L > 0$ . The desert  $(-\gamma_1, \gamma_1)$  and the gaps longer than the Nyquist step  $2\pi/L$  fix the *exponential* smallness (*sampling-floor* Prop. 2, geometric law of *depth-geometry-quorum-mechanism*). Beurling’s  $D^- >$  is too coarse to see  $e^{-L\gamma_1}$ .

## 2 Unconditional identity: Weil, not a real Gram

Without RH the left side  $Q_L$  is still the prime-side pairing (pole + archimedean – towers  $n \leq e^L$ ). The right side is

$$Q_L(f) = \sum_{\rho} \hat{g}_f(\rho),$$

sum over *all* nontrivial zeros  $\rho$ , with  $\hat{g}_f$  the entire function attached to  $f$  by the explicit formula (Weil–Bombieri normalization of the repository, App. B of *quorum-theorem*). On the line,  $\rho = \frac{1}{2} + i\gamma$  and  $\hat{g}_f(\rho) = F(\gamma)^2$  (even convention of the window). Off the line,  $\rho = \frac{1}{2} + \sigma + i\gamma$  with  $\sigma \neq 0$ , the term is not a square of a real sample. The form is no longer a Gram of evaluators on  $\mathbb{R}$ .

*Remark.* Global Weil positivity —  $Q(h) \geq 0$  for every admissible  $h$  on  $\mathbb{R}$ , not just on one  $W_L$  — is equivalent to RH. That is Weil’s theorem. Restricting the test class to  $W_L$  loses the equivalence: a form that is positive on every compact window is a stronger statement than positivity on one window, and is again RH.

### 3 Visibility of an off-line zero

Let  $\rho = \frac{1}{2} + \sigma + i\gamma$  with  $\sigma \neq 0$ . The contribution of the pair  $(\rho, 1 - \rho)$  to  $Q_L$  is  $2\operatorname{Re} \hat{g}_f(\rho)$ . A test function whose spectral mass sits at  $\rho$  can make this term dominate and negative.

**Lemma 1** (visibility, schematic). *If  $F \in PW_\tau$  can be chosen with  $|F(\sigma + i\gamma)| \geq c\|F\|$  and  $\sum_{t \in \mathbb{R}} |F(t)|^2$  small compared with that, then  $Q_L$  takes a negative value as soon as the off-line term is included. Such an  $F$  exists in  $PW_\tau$  once  $\tau$  exceeds a constant times  $|\sigma| + 1$  and the distance from  $\gamma$  to the real sampling nodes is not large compared with  $\tau^{-1}$ .*

This is Paley–Wiener, not arithmetic: an entire function of type  $\tau$  cannot peak far off the real axis unless  $\tau$  is large enough (Phragmén–Lindelöf / Bernstein). Consequence for the repository: positivity of  $Q_L$  at  $\mu = e^L = 11$  is compatible with an off-line zero at height  $10^6$ . It is not compatible with an off-line zero inside the band of  $V_N$  once  $N$  resolves that height. A single certified window does not force RH; a uniform statement over  $L \rightarrow \infty$  would.

**Wrong continuation.** Continuing the Dirichlet-kernel formula  $2\sin(sL/2)/(s\sqrt{L})$  to  $s = \sigma + i\gamma$  explodes as  $\sinh(\gamma L/2)$  and produces  $\lambda_{\min} \sim -10^7$  already at  $\sigma = 0.05$ . That is the Fourier transform in a complex *frequency*, not the Weil pairing. Discarded.

**Weil weight.** The pairing at  $\rho = \frac{1}{2} + \sigma + i\gamma$  is  $\hat{\eta}(\rho) = \int_{-L/2}^{L/2} \eta(x) e^{\sigma x} e^{i\gamma x} dx$ , bounded by  $e^{|\sigma|\tau} \sqrt{L}$ . At  $\sigma = 0$  it recovers the on-line formula. The quadratic term attached to the pair  $(\rho, 1 - \rho)$  is the Gram of  $(\hat{\eta}_n(\rho)\hat{\eta}_m(1 - \rho))$ . On  $V_9$  at  $\mu = 11$ , forty real zeros already in  $Q_{\text{line}}$ :

| $\sigma$ at $\gamma_1$ | $\lambda_{\min}$ of $Q_{\text{line}} + F(\rho)F(1 - \rho)$ | of $Q_{\text{line}} +  F(\rho) ^2$ |
|------------------------|------------------------------------------------------------|------------------------------------|
| 0                      | $+3 \times 10^{-22}$                                       | $+3 \times 10^{-22}$               |
| 0.01                   | $-5.4 \times 10^{-5}$                                      | $+3 \times 10^{-17}$               |
| 0.10                   | $-7.3 \times 10^{-3}$                                      | $+4 \times 10^{-17}$               |
| 0.50                   | $-0.23$                                                    | $+4 \times 10^{-17}$               |
| 1.00                   | $-1.28$                                                    | $+3 \times 10^{-17}$               |

$|F(\rho)|^2$  never crosses.  $F(\rho)F(1 - \rho)$  crosses at once, at height  $\gamma_1$ , with size  $O(\sigma^2)$  at small  $\sigma$  — not  $10^7$ . The same extra term at height 80 or 200 leaves  $\lambda_{\min}$  at  $10^{-22}$ : this window does not see a high off-line zero. Visibility is local in height, mild in  $\sigma$ , and tied to  $\gamma_1$ . A certified  $Q_{11} \succ 0$  still allows an off-line zero at large height. It does not allow one at  $\gamma_1$  if the pairing really is  $F(\rho)F(1 - \rho)$ . That last “if” is the remaining identification with App. B of *quorum-theorem*.

### 4 What de Branges supplies, and what it does not

A de Branges space  $\mathcal{H}(E)$  is a Hilbert space of entire functions with a reproducing kernel built from a Hermite–Biehler function  $E$ . Paley–Wiener  $PW_\tau$  is the model case  $E(z) = e^{-i\tau z}$ . Sampling and interpolation in  $\mathcal{H}(E)$  are completely described (de Branges, Beurling–Malliavin for the PW case): a set is sampling when its density exceeds the type, with a structure condition on the gaps.

Two precise contacts with  $Q_L$ .

**PW as  $\mathcal{H}(E)$ .** Theorem 1 is the PW-special case of “the zeros sample  $\mathcal{H}(E)$ ”. The desert bound is the reproducing-kernel mass of the hole  $(-\gamma_1, \gamma_1)$  in that space — Slepian is the first eigenfunction of the projection onto the hole. Sub-Nyquist gaps are the same statement for a union of holes. The geometric law is a conjecture about the bottom eigenvalue of a Gram of reproducing kernels at the zeros. de Branges gives the language; it does not give  $a = 1.69$ ,  $b = 0.82$ .

**The de Branges approach to  $\xi$ .** There is a different  $E$ , built from  $\xi$ , whose space  $\mathcal{H}(E)$  has the zeros of  $\xi$  as a distinguished set. RH is equivalent to axioms on that  $E$  (de Branges 1986 and later variants). That  $E$  is *not*  $e^{-i\tau z}$ . Importing a theorem about  $PW_\tau$  into the  $\xi$ -space is the usual gap in the programme. The windowed form  $Q_L$  lives in PW. The  $\xi$ -space lives on the critical strip. They share a philosophy (positivity  $\Leftrightarrow$  sampling on a line) and not a kernel.

## 5 What this route can still do in the repository

1. State Theorem 1 as the definition of the floor, not as a numerical finding. Already done in *sampling-floor*.
2. Replace “hyper-nullity” language by “the sampling constant of a set with a hole of width  $\gamma_1$  and type  $\tau$ ”. Same numbers, right theory.
3. Prove a quantitative visibility lemma: an explicit  $L(\sigma, \gamma)$  beyond which an off-line pair at  $\frac{1}{2} + \sigma + i\gamma$  forces  $\lambda_{\min}(Q_L) < 0$ . This is analysis, not flint.
4. Compute, on  $V_N$ , the Gram with one complex pair added, and record the  $L$  at which  $\lambda_{\min}$  crosses. A measurement of the schematic lemma. Not RH.
5. Do *not* claim that  $c_L > 0$  for all  $L$  unconditionally. That statement is RH.

## Status

Theorem 1 (under RH) is already in the living corpus. The present note adds the unconditional reading, the visibility obstruction, and the boundary between PW-de Branges and  $\xi$ -de Branges. No new theorem. No movement of RH.